

Advanced topics in seasonal adjustment: JDemetra+: Additional Tools

Thomas Witthohn, Jan Gerhardt / DG Statistics
4 November 2025

■ Agenda

- Massproduction in UI
- Cruncher
- Plug-in Multidoc
- Plug-in XLSX2WS

■ Annex

- Plug-in Dataprovider Random
- Anomaly Detection

Mass production

Motivation

Daily business

Mass production

Motivation

Daily business

- lots of data

Daily business

- lots of data
- time constraints

Daily business

- lots of data
- time constraints
- production chain

Daily business

- lots of data
- time constraints
- production chain
- → Time consuming for exporting data manually

JD+ embedded solution for multi-documents

- Run multi-document
- Go to [Menu](#) → [Name of selected multi-doc panel](#) → [Output...](#)
- Select output source(s)
- Configure output
- Save output

Configure Output source

- CSV
- CSV matrix
- Excel
- Txt

Configure Output series: y , t , s_a , s , i , y_{cal}

Configure items: statistics of the series

Mass production

Using the App III

Batch output

Output + -

File icons

Csv

Csv matrix

Excel

Txt

Location

folder C:\Daten\Temp

File Prefix series

Layout

presentation List

Full series name ☒

Content

series [y, t, sa, s, i, ycal]

Csv

OK Cancel

Useful links - Using the App

Repository (github/jdemetra)



`https://github.com/jdemetra/jdemetra-app/releases/latest`

Documentation (github.i.o)



`...JDemetra-documentation/case studies/output`

New Documentation (in development)



`https://.../jdemetra-new-documentation/gui/output`

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Batch Processing

Cruncher (I/V)

Console tool that reestimates all the multi-documents defined in a JD+ workspace.

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- copy [jwsacruncher-2.2.6.zip](#) to [C:\Workspace\R](#)

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- executable: ["...\\jwsacruncher-2.2.6-bin\\bin\\jwsacruncher"](#)

Batch Processing

Cruncher (II/V)

Open cmd (command prompt window "Eingabeaufforderung")

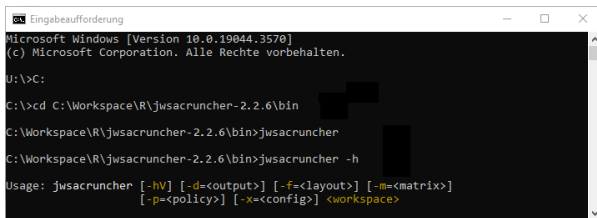
Batch Processing

Cruncher (II/V)

Open cmd (command prompt window "Eingabeaufforderung")

Help

Jwsacruncher -h



```
Microsoft Windows [Version 10.0.19044.3570]
(c) Microsoft Corporation. Alle Rechte vorbehalten.

U:\>C:

C:\>cd C:\Workspace\R\jwsacruncher-2.2.6\bin

C:\Workspace\R\jwsacruncher-2.2.6\bin>jwsacruncher

C:\Workspace\R\jwsacruncher-2.2.6\bin>jwsacruncher -h

Usage: jwsacruncher [-hV] [-d=<output>] [-f=<layout>] [-m=<matrix>]
                  [-p=<policy>] [-x=<config>] <workspace>
```

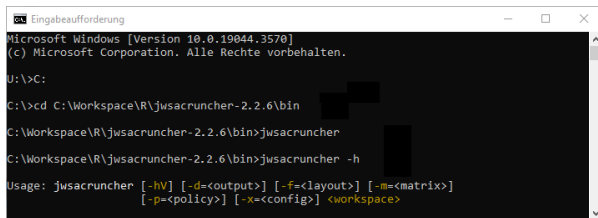
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```

Create default params file

Jwsacruncher

Parameter file (*.params) for configuration of the csv output:

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 - $\rightarrow$ sort order of coefficients, parameters etc. in csv output

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- csv separator $\leadsto$ `csvseparator=" ; "`
- number of decimals $\leadsto$ `ndecs="6"`
- refreshing policy $\leadsto$ `<policy>`
- diagnostics $\leadsto$ `<matrix>`
 - $\rightarrow$ sort order of coefficients, parameters etc. in csv output
- time series $\leadsto$ `<tsmatrix>`
 - $\rightarrow$ For SA with X-13: Decomposition files for A1, ..., B1, ..., C1, ...
 - $\rightarrow$ Remove files containing a "?"-proxy

Refreshing policy <policy>

Option JD+	Option Cruncher
Current adjustment	current
Fixed model	fixed
Re-estimate regression coefficients	fixedparameters
+ Arima parameters	parameters
+ Last Outliers	lastoutliers
+ All Outliers	outliers
+ Arima model	stochastic
Concurrent adjustment	complete or concurrent

Refresh and process all multi-documents of the workspace [Trade.xml](#)

Batch Processing

Cruncher (V/V)

Refresh and process all multi-documents of the workspace [Trade.xml](#)

Refresh Workspace

```
Jwsacruncher C:\Workspace\R\Trade.xml -x
```

```
C:\Workspace\R\jwsacruncher-2.2.6\bin\wsacruncher.params
```

Refresh and process all multi-documents of the workspace [Trade.xml](#)

Refresh Workspace

```
Jwsacruncher C:\Workspace\R\Trade.xml -x  
C:\Workspace\R\jwsacruncher-2.2.6\bin\wsacruncher.params
```

(Default) Output files

C:\Workspace\R\Trade\Output

Batch Processing

Cruncher (V/V)

Refresh and process all multi-documents of the workspace [Trade.xml](#)

Refresh Workspace

```
Jwsacruncher C:\Workspace\R\Trade.xml -x  
C:\Workspace\R\jwsacruncher-2.2.6\bin\wsacruncher.params
```

(Default) Output files

```
C:\Workspace\R\Trade\Output
```

Indicators

```
C:\Workspace\R\Trade\Output\SAProcessing-1\demetra_m.csv
```


Cruncher

 <https://github.com/jdemetra/jwsacruncher/releases>

 <https://github.com/jdemetra/jwsacruncher/wiki>

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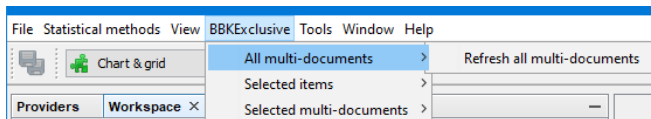
- Streamline the daily workflow of Bundesbank

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- Reduce error potential

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- Reduce error potential
- Reports for internal revision processes

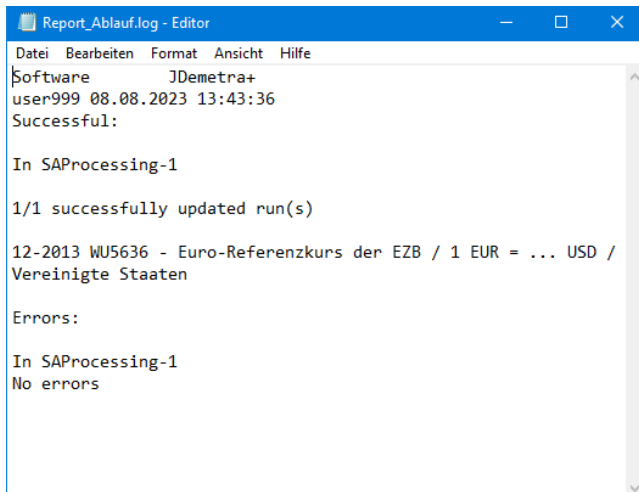
MultidocExtension

Introduced features I



MultidocExtension

Introduced features II



The screenshot shows a text editor window with a blue title bar containing the text 'Report_Ablauf.log - Editor' and standard window controls. The editor has a menu bar with 'Datei', 'Bearbeiten', 'Format', 'Ansicht', and 'Hilfe'. The text area contains the following log entries:

```
Software          JDemetra+  
user999 08.08.2023 13:43:36  
Successful:  
  
In SAProcessing-1  
  
1/1 successfully updated run(s)  
  
12-2013 WU5636 - Euro-Referenzkurs der EZB / 1 EUR = ... USD /  
Vereinigte Staaten  
  
Errors:  
  
In SAProcessing-1  
No errors
```

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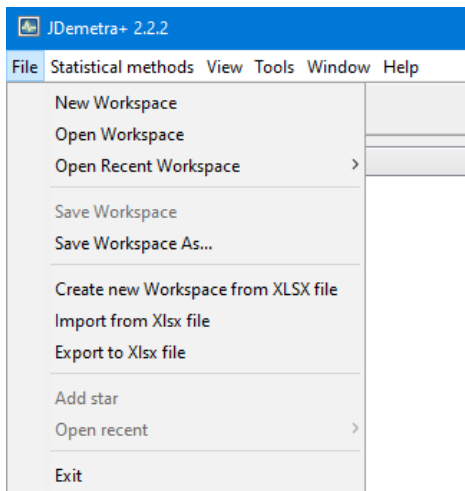
- Change the same specification in several items

- Change the same specification in several items
- Exchange of workspaces without the data

- Change the same specification in several items
- Exchange of workspaces without the data
- Send proposed changes of items to colleagues

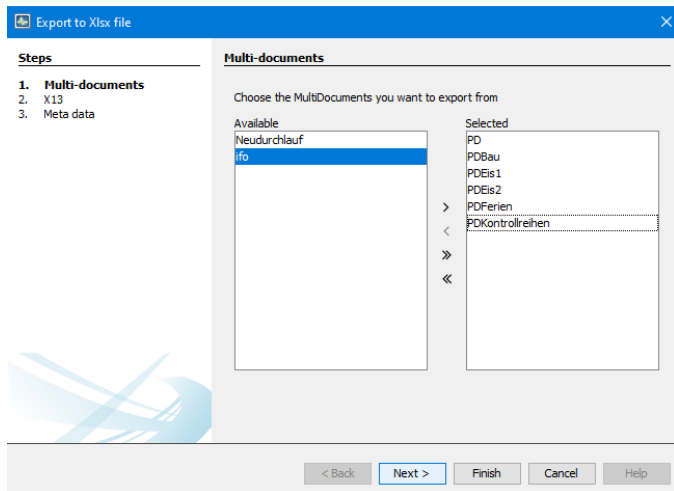
Xlsx2Ws

Using the Plug-In I



Xlsx2Ws

Using the Plug-In II



- Prepare a new version for JDemetra Version 3
- Make more providers available for easy changes
- Write templates/documentation for implementing internal providers

Requirements

- Java 17 or higher

Issues and Code on GitHub

- <https://github.com/bbkrd/Xlsx2Ws>

Documentation

- <https://bbkrd.github.io/pages/xlsx2ws/>

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Random ARIMA generator

Motivation

- No time series (provider) available
- Anonymous data
- Testing new features

Random ARIMA generator

tutorial-plugin-random-2.0.0.nbm

A plugin that adds a new time series provider. This provider generates pseudo-random time series using ARIMA.

The random numbers are generated with the Xorshift random number generator

Random ARIMA generator

ARIMA:

Regular auto-regressive order (p)	max. 3
Regular differencing order (d)	max. 2
Regular moving average order (q)	max. 3
Seasonal auto-regressive order (bp)	max. 1
Seasonal differencing order (bd)	max. 1
Seasonal moving average order (bq)	max. 1
coeff	
Stde of coeff	max. 1

Random ARIMA generator

Other:

Random seed	input for random variable generator
Frequency	4 or 12
Number of periods	
Number of series	

Useful links - Random Data Generator

Tutorial-plugin-random:



<https://github.com/nbbird/jdemetra-tutorial>

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The internal library of JD+ is extremely powerful. Could we use it for other analyses? **Questions:**

- Has a time series outlier?
- Are the last n-values in the range of expectation?

| Anomaly detection

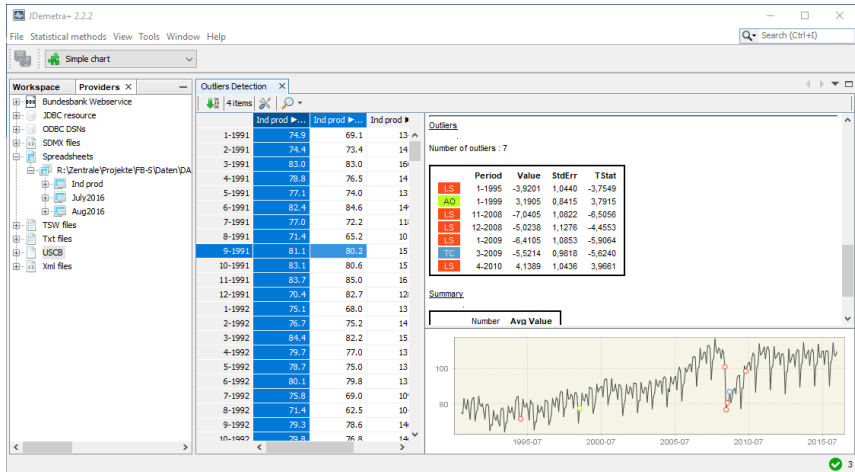
Internal Plug-in: NbDemetra Anomaly detection

The task of this Plug-in is to identify outliers of a time series:

- 1 Detection of outliers
- 2 Check last

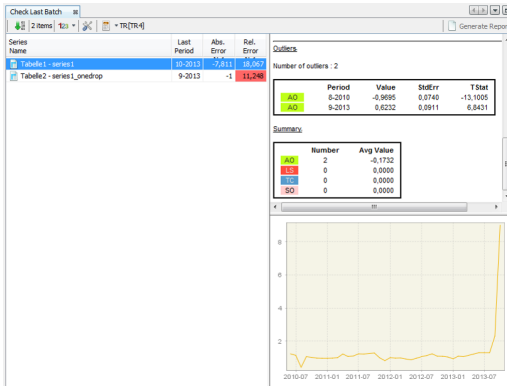
Anomaly detection

Outliers Detection



Anomaly detection

Check Last I



A list of time series can be processed

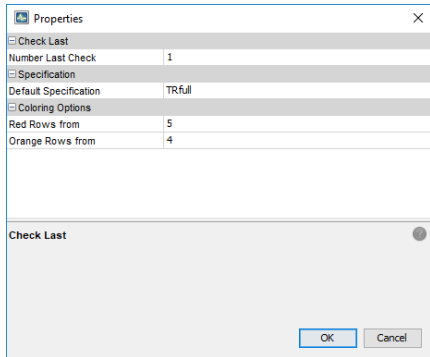
The last (1,2,3) observation is compared with its forecasted value

If the difference is too high compared with the volatility of the series the value is marked red

The estimated ARIMA model and some diagnostics are reported

Anomaly detection

Check Last II



The screenshot shows a 'Properties' dialog box with a close button (X) in the top right corner. The dialog is divided into three sections: 'Check Last', 'Specification', and 'Coloring Options'. The 'Check Last' section contains a table with one row: 'Number Last Check' with the value '1'. The 'Specification' section contains a table with one row: 'Default Specification' with the value 'TRfull'. The 'Coloring Options' section contains two rows: 'Red Rows from' with the value '5' and 'Orange Rows from' with the value '4'. At the bottom of the dialog, there are 'OK' and 'Cancel' buttons.

Check Last	
Number Last Check	1

Specification	
Default Specification	TRfull

Coloring Options	
Red Rows from	5
Orange Rows from	4

Check Last

OK Cancel

The modelling from TRAMO is used to detect the different types of outliers

The configurations are the same as the ones in the modelling part of the workspace

Method

- Calculation of the selected ARIMA model
- Calculate forecast f_1 and the forecast innovations ε_1 with Kalman filter
- Calculate the absolute error $|f_1 - y_n|$ and relative error $|\frac{f_1 - y_n}{\varepsilon_1}|$
- Compare the relative errors with the threshold (4,5)

Useful links - Anomaly detection

[Documentation \(github.io\)](#)



`...JDemetra-documentation/Basic time series analysis`